

Conceptual Answers – Order Flow Imbalance (OFI) Task

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1. Motivation Behind Measuring OFI at Multiple Depth Levels

Order Flow Imbalance (OFI) captures the net pressure between buyers and sellers. Measuring it only at the best level (Level 1) may miss hidden liquidity deeper in the book. By extending OFI to multiple depth levels (e.g., top 5 or top 10), we can better observe passive buying/selling interest that could influence short-term price dynamics.

As shown in the paper *Cross-Impact of Order Flow Imbalance in Equity Markets*, multi-level OFI offers more robust predictive power and reflects structural pressure in the market that Level 1 alone cannot capture.

2. Why Use LASSO Instead of OLS for Cross-Impact Estimation

In cross-impact modeling, where a stock's return may depend on the OFIs of many other assets, Ordinary Least Squares (OLS) becomes unreliable due to multicollinearity and high dimensionality.

LASSO (Least Absolute Shrinkage and Selection Operator) addresses these problems by:

- Performing variable selection (shrinking irrelevant coefficients to zero)
- Handling situations where the number of predictors exceeds the number of observations
- Producing sparser and more interpretable models

This makes LASSO ideal for estimating cross-impact coefficients where not all assets are equally influential.

3. Why OFI is a Better Predictor of Short-Term Returns Than Trade Volume

Trade volume is a symmetric measure that aggregates buyer- and seller-initiated trades without indicating direction. It cannot distinguish whether the market is being driven by aggressive buying or selling.

In contrast, OFI captures the *directional pressure* between bid and ask sides by measuring changes in order book sizes. This makes OFI a more direct indicator of imbalance and potential price movement.

Empirical evidence in the paper shows that OFI explains a larger portion of short-term return variance compared to volume, particularly when constructed across multiple depth levels or integrated using PCA.